

Critical Benjamin–Ono–BBM Quasi-Invariance: A Literature Answer to arXiv:2103.04408

Automated mathematics run `fa_banach_001`, lane 7

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Status. Literature already answered: the core quasi-invariance statement at critical dispersion is proved in arXiv:2107.03445. The exact density formula and the stronger all-finite- p integrability conclusion remain open.

Source future-work statement

Genovese, Lucà, and Tzvetkov study the fractional BBM equation

$$\partial_t u + \partial_t |D_x|^\beta u + \partial_x u + \partial_x(u^2) = 0.$$

Their Theorem 1.3 treats $\beta \in (1, 2]$ and proves quasi-invariance of exponentially cut-off Gaussian measures, all-finite- p integrability of the transported densities, and (under an additional regularity hypothesis) an explicit density formula. Immediately afterward, on PDF page 4, they write:

It is however known that (1.4) is globally well-posed for $\beta = 1 \dots$ and we plan to study the extension of Theorem 1.3 to the case $\beta = 1$ in a future work.

This is the sole extracted future-work signal in arXiv:2103.04408.

Decisive follow-up

The same authors' later paper arXiv:2107.03445 is devoted precisely to the $\beta = 1$ periodic Benjamin–Ono–BBM equation. Its introduction explicitly states that it extends the previous $\beta > 1$ achievements to the minimal dispersion case $\beta = 1$. Theorem 1.1 (PDF page 2) proves that for every $s > 1$ and energy cutoff $R > 0$, the cut-off Gaussian measure $\tilde{\gamma}_s$ is quasi-invariant under the flow and, for each time t , its transported density belongs to $L^{p(t,R)}$ for some $p(t, R) > 1$. Remark 1.2 further states that the energy cutoff can be removed when only quasi-invariance of the reference Gaussian measure γ_s is required.

The supporting authors therefore knew they were answering the critical dispersion continuation: the equation, parameter endpoint, authors, and claimed extension are explicit matches.

Scope limitation

This is a complete literature answer to the *core quasi-invariance* part of the future-work target, not to every clause of the earlier Theorem 1.3. The follow-up explicitly says that the minimal dispersion prevents its method from identifying the Radon–Nikodym density; that question remains open. It also proves membership in some L^p , $p > 1$, rather than in every finite L^p . Moreover, it works with an energy cutoff rather than the earlier exponential H^s cutoff, although Remark 1.2 yields the uncut quasi-invariance conclusion.

Search evidence

Run-index searches for the exact arXiv id, title, equation, and critical parameter found no prior packet. An exact-title arXiv query located arXiv:2107.03445v2. The supporting abstract, introduction, Theorem 1.1, and Remark 1.2 were checked against the source’s Theorem 1.3 and future-work sentence. The match is explicit rather than an agent-inferred reformulation.

References

- [1] G. Genovese, R. Lucà, and N. Tzvetkov, *Transport of Gaussian measures with exponential cut-off for Hamiltonian PDEs*, arXiv:2103.04408v3; Ann. Inst. H. Poincaré C Anal. Non Linéaire 39 (2022), 1465–1503.
- [2] G. Genovese, R. Lucà, and N. Tzvetkov, *Quasi-invariance of Gaussian measures for the periodic Benjamin–Ono–BBM equation*, arXiv:2107.03445v2; Stoch. Partial Differ. Equ. Anal. Comput. 10 (2022), 1482–1507.